

Muhammad Ifte Khairul Islam

Atlanta, GA — 📞 405-780-2169 — ✉ iftekhairulshuvo@gmail.com — 🔗 [linkedin.com/in/ifte-islam-909](https://www.linkedin.com/in/ifte-islam-909) — 🐙 github.com/iftekhairul

CAREER ACCOMPLISHMENTS

- Course Instructor for Data Structures (100+ students); designed lectures, labs, and exams with strong student feedback.
- GTA for Machine Learning, Network Science, and Software Development; supported labs and student learning.
- Mentored undergraduate and graduate students on graph mining and machine learning projects.
- Lead author of multiple publications in leading conferences including PAKDD, CIKM, and IEEE Big Data.
- Recipient of multiple research awards, including the Outstanding Graduate Student Research Award (OSU, 2021).

TEACHING INTEREST

Introduction to Programming, Data Structures and Algorithms, Discrete Mathematics, Software Engineering, Database Systems, Machine Learning, Artificial Intelligence, Deep Learning,

RESEARCH INTEREST

Graph Mining, Graph Machine Learning, Graph Neural Network, Applied Machine Learning, Text Mining, Social Network Analysis, Bioinformatics, BioMedicine, Health Data Analytics

EDUCATION

PhD in Computer Science

Georgia State University (Transferred from OSU)

Thesis: Mining Local and Global Structure on Static and Dynamic Graphs

Spring 2023 – Summer 2025

Atlanta, GA

PhD in Computer Science

Oklahoma State University (Transferred to GSU)

Fall 2019 – Fall 2022

Stillwater, GA

Bachelor of Science in Computer Science and Engineering

Daffodil International University

Jan, 2014 - Dec, 2017

Dhaka, Bangladesh

TEACHING EXPERIENCE

Course Instructor – Data Structures

Georgia State University

Fall 2024

- Designed and prepared course materials including syllabus, lecture notes, programming labs, and exams.
- Delivered lectures and facilitated active learning for 100+ undergraduate students.
- Supervised and coordinated a team of 3 teaching assistants, overseeing grading and lab instruction.
- Integrated hands-on coding projects to reinforce algorithms and data structures.
- Received strongly positive student evaluations, highlighting clear instruction and supportive teaching style.

Graduate Teaching Assistant

Georgia State University

Spring 2023 – Summer 2025

- Supported instruction and grading for courses in Machine Learning, Software Development, and Network Science.
- Lab Instructor for Principles of Computer Science, teaching Python programming fundamentals.
- Guest Lecturer in Software Development (OOP Design Patterns, SOLID principles).
- Guest Lecturer in Network Science (Graph Neural Networks and graph classification).
- Mentored students on programming assignments and research projects.

Graduate Teaching Assistant

Oklahoma State University

Fall 2019 – Fall 2022

- TA for Data Structures and Algorithms, Operating Systems, Computer Architecture, Database Systems, and C++ Programming.

- Graded assignments, projects, and exams while ensuring timely feedback to students.
- Assisted students with lab assignments and provided academic support during office hours.
- Supported instructors in course administration and facilitated classroom learning.

RESEARCH EXPERIENCE

Graduate Research Assistant

Jan 2023 – Aug 2025

Georgia State University

- Proposed **Dynamic Graph Contrastive Learning (DyGCL)**, a framework combining local and global graph encoders with contrastive learning and attention to predict dynamic events (e.g., political unrest, disease outbreaks, criminal activities); published in *IEEE Big Data* 2024.
- Developed a **truss-based graph sparsification model** to prune edges from dense regions, improving efficiency of GNNs.
- Introduced **Graph Contrastive Learning with Graph Compression (GCLC)** using k-core and k-truss decompositions for structure-aware view generation; paper currently under review.
- Contributed to **Seq-HyGAN**, a hypergraph attention network for sequence classification; published in *CIKM* 2023.
- Conducted extensive experiments on real-world datasets (social and biological networks), achieving improvements over state-of-the-art baselines.

Graduate Research Assistant

Aug 2019 – Dec 2022

Oklahoma State University

- Developed **Motif-based Graph Pooling (MPool)**, a multi-channel framework combining selection- and clustering-based pooling; published in *PAKDD* 2023.
- Developed **Network Embedding with Compression and Learning (NECL)**, a graph compression-based embedding framework; published as first author in *Frontiers in Big Data*, 2021.
- Co-authored **HIN-DDI**, a heterogeneous information network model for drug-drug interaction prediction; published in *CIBCB*.
- Collaborated on a **graph-based model for drug abuse event detection from Twitter**, applying GNNs to text graphs; published in *IEEE Big Data*.

INDUSTRY EXPERIENCE

Software Engineer Intern

Dec 2017 – Jul 2019

US Software Ltd., Dhaka, Bangladesh

- Designed and developed enterprise applications using **Java, Spring, and MySQL**.
- Implemented core modules, including authentication, reporting, and data management features.
- Collaborated with cross-functional teams to deliver scalable and maintainable software solutions.
- Applied software engineering best practices such as version control (Git), testing, and documentation.

PUBLICATIONS

1. **Islam, M. I. K.**, Saifuddin, K. M., Hossain, T., & Akbas, E. “DyGCL: Dynamic graph contrastive learning for event prediction.” In *IEEE International Conference on Big Data (BigData)*, pp. 559–568, 2024.
2. **Islam, M. I. K.**, Tanvir, F., Johnson, G., Akbas, E., & Aktas, M. E. “Proximity-based compression for network embedding.” *Frontiers in Big Data*, vol. 3, p. 608043, 2021.
3. **Islam, M. I. K.**, Khanov, M., & Akbas, E. “MPool: Motif-based graph pooling.” In *Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)*, pp. 105–117, 2023.
4. Saifuddin, K. M., May, C., Tanvir, F., **Islam, M. I. K.**, & Akbas, E. “Seq-HyGAN: Sequence classification via hypergraph attention network.” In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM)*, pp. 2167–2177, 2023.

5. Tanvir, F., Saifuddin, K. M., **Islam, M. I. K.**, & Akbas, E. "Predicting drug-drug interactions using heterogeneous graph attention networks." In *Proceedings of the 14th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, pp. 1–6, 2023.
6. Tanvir, F., Saifuddin, K. M., **Islam, M. I. K.**, & Akbas, E. "DDI prediction with heterogeneous information network-meta-path based approach." *IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB)*, vol. 21, no. 5, pp. 1168–1179, 2024.
7. Hossain, T., Saifuddin, K. M., **Islam, M. I. K.**, Tanvir, F., & Akbas, E. "Tackling oversmoothing in GNN via graph sparsification: A truss-based approach." *ECML PKDD*, pp. 161–179, 2024.

Full list of publications available on [Google Scholar Profile](#)

AWARDS

- **NSF Student Travel Award**, IEEE BigData Conference (2024) and ACM KDD (2023).
- **Outstanding Computer Science Researcher Award** (2021), Department of Computer Science, Oklahoma State University.
- **Computer Science Fellowship**, Department of Computer Science, Oklahoma State University.

MENTORING

- Mentored undergraduate students through the **NSF REU Program** (2023); guided a project on *Graph Compression using Contrastive Learning*. – Outcome: Student published a paper at the *2024 International Conference on Computer Communications and Networks (ICCCN)*.
- Supervised Master's students for their thesis research on graph mining and machine learning.
- Provided guidance on project design, coding implementation, and paper writing, fostering student growth in research and technical communication.

SKILLS

- **Programming:** Python (Advanced), Java (Intermediate), C (Intermediate)
- **Machine Learning & Deep Learning:** PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn
- **Data Science & Libraries:** Pandas, NumPy, SciPy, Matplotlib, NetworkX
- **Tools & Platforms:** Git, Linux, Jupyter Notebook, LaTeX

PROFESSIONAL SERVICES

- **Conference Reviewer:** KDD 2024, KDD-UG 2022, ECML-PKDD-2024
- **Conference Sub-reviewer:** PAKDD 2023, Complex Networks 2021
- **Industrial Collaborator:** DIU Computer Programming Club (CPC), 2016

REFERENCES

- **Dr. Esra Akbas**
Associate Professor
Department of Computer Science
Georgia State University, Atlanta, GA
Email: eakbas1@gsu.edu
- **Dr. Khaled Mohammed Saifuddin**
Assistant Professor
Department of Computer Science
Southern Illinois University Carbondale
Email: khaledmohammed.saifuddin@siu.edu
- **Dr. S. M. Towhidul Islam**
Lecturer
Department of Computer Science
Georgia State University, Atlanta, GA
Email: sislam9@gsu.edu